

PhoneScan - Integration Manual

1. Introduction

This document serves as a technical reference for integrating software with the PhoneScan device, a USB barcode/QR scanning device.

PhoneScan is designed as a composite USB device providing both human input emulation and a control interface for configuration and management.

2. USB Device Properties

Device Identification

- VID: 1209
- PID: C55C
- Product Name: PhoneScan

USB Interfaces

The device enumerates as a USB Composite Device with the following interfaces:

1. CDC-ACM (Virtual Serial Port)
 - Used for configuration and device control
 - Command-based ASCII interface
2. HID Keyboard Interface
 - Used for output of scanned barcode/QR data
 - Emulates a standard USB keyboard

3. Serial (CDC-ACM) Interface

Overview

The CDC interface provides a command-based control channel for configuring scanner behavior and querying device state. • Baud Rate: Dynamic (no strict requirement due to USB CDC abstraction)

- Recommended Baud Rate: 57600
- Encoding: ASCII
- Line Ending: \n or \r

3.1 Command Format

- Commands are case-insensitive
- Commands must be terminated with newline (\n) or carriage return (\r)
- Invalid commands return an error string

3.2 Supported Commands

Command	Description
next	Cycle to next scanning mode
manual	Set Manual Mode (trigger-based scanning)
sense	Set Sense Mode (auto detection)
continuous	Set Continuous scanning mode
mode	Print current scanning mode
status	Print full device status
sleep	Disable scanner power
wake	Enable scanner power
trig	Manually trigger a scan pulse
bootloader	Reboot device into bootloader mode
debugmode	Debug mode reading
help	List all supported commands

3.3 Example Command Usage

Query Mode

mode

Response

```
[ mode ] Sense
```

Status Query

status

Response

```
+-----+
| mode      : Continuous
| sleeping  : no
+-----+
```

Invalid Command

foo

Response

```
[ err ] unknown: foo
        type 'help' for commands
```

4. HID Keyboard Interface

Overview

The HID interface outputs scanned data as keyboard keystrokes.

Behavior

- Each scanned character is transmitted as a HID key event
- Supports full ASCII character set:
 - a–z, A–Z
 - 0–9
 - Common symbols (- _ = + [] \ | ; : ' " , . / ? ! @ # \$ % ^ & * ()
- Uppercase characters are sent using SHIFT modifier
- No automatic newline is appended unless present in scanned data

Output Format

Example barcode:

ABC123

Appears on host as a string of individual HID key events:

ABC123

Also the text [scan] comes over Serial before each successful decode.

5. Scanning Modes

5.1 Manual Mode

- Scanning only occurs when explicitly triggered via `trig` command.

5.2 Sense Mode

- Automatically detects and scans barcodes when present.

5.3 Continuous Mode

- Scanner is always active.

6. System Behavior Notes

- CDC and HID interfaces operate independently
- Barcode output behaves exactly like a physical USB keyboard
- Command latency is typically < 100 ms
- Scanner may briefly flush UART buffer during command execution
- No flow control (RTS/CTS) is implemented on CDC
- Device initializes with 2-second boot delay
- Mode changes are automatically saved to flash